

Literature Review – GECCO

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Key words:

rural	Sweden	barriers
urban	Europe	drivers
	Global North	solution

CE = community energy

EC = energy community

RE = renewable energy

RES = renewable energy sources

REC = renewable energy community

ARTICLES

1.1. STREIMIKIENE, ET AL., 2021

Streimikiene, D., Baležentis, T., Volkov, A., Morkunas, M., Žickiene, A. and Streimikis, J, 2021. Barriers and Drivers of Renewable Energy penetration in Rural Areas. *energies*, [e-journal] 14(6452). <https://doi.org/10.3390/en14206452>.

- Energy inequality and lack of agency for certain energy citizenship can be found in particular places. The possibility for energy citizenship was enabled in some places which disabled it for others (p.3).
- Renewable energy initiatives in rural communities can have several benefits for the farmers, like the additional income or a self-sufficient supply of electricity. However, the lack of knowledge is a hindrance for the realization of such projects (p.6).
- **Socio-political and regulatory barriers:**
 - “The main reason RES technologies are not being implemented faster is the regulations that protect big fossil energy producers from free competition due to their high lobbying power.” (p.10).
 - Low income and high electricity prices as well as unstable political support stagnate renewable energy initiatives. Rural areas are more prone to poverty and higher inequality levels (p.10)
 - Insufficient local community engagement and participation (p.10)
 - Ineffectiveness of regional governance due to dependence on the national level (p.10)
 - High requirement for planning permission of RES technologies (p.10)
 - Agricultural practices (machinery, pest management, irrigation etc.) are very energy-intensive and the most productive approach would be the electrification of these processes. However, there is no to little knowledge about how to use renewable electricity sources for the same practices and the right strategies, business models, financial and other resources as well as policies and regulations do not exist or are not in place yet to make the transformation possible on a larger scale (pp.4-6).
- **Behavioural and psychological barriers:**
 - Worries about the complexity as well as the lack of knowledge about advantages of RES technologies (p.10)
 - Fears about efficiency and safety (p.10)
- **Environmental barriers:**
 - Big renewable energy projects often face resistance from farmers in rural areas due to negative impacts on the environment, for example on soil temperature and/or noise (p.6).
- Policies and measures as part of the solution to overcome barriers of renewable energy initiatives in rural areas (p.13).
- “If the rural community has a political motivation that is similar to the governmental institutions, this provides many openings for rural community renewable energy initiatives.” (p.21).

- Non-economic policies are more powerful, due to the affect it can have on energy consumer's attitude towards using RES technologies (p.22).

rural Europe barriers solution

1.2. ERIKSSON BERGGREN, ET AL., 2023

Eriksson Berggren, S., Witt, T., Van der Linden, E., Saes, L., Edander Arvefjord, L., Heckenberg, D., Iglauer, T., Sutinen, L., Hanning, E. and Melin, G., 2023. *Energy Communities*. [online] Nordic Energy Research. Available at: <<https://pub.norden.org/nordicenergyresearch2023-03/index.html?fbclid=IwAR1gksRnX2iC9RxtDyc8COuIJCAQGu6kMzxa38BA5R5F984RIEXX3u9sC8>> [Accessed 27 September 2023].

Summary: Exchange of views on lessons learned and common experiences in the Nordic and European countries. A desk research and literature review that also led to studies of the following countries: Denmark, Sweden, Finland, Norway, Germany, Austria, Netherlands.

Overall conclusions:

- The studied countries rarely use the EU legal definitions of energy community (p.7) and has not been transposed into national legislations in most countries – Sweden included (p.25) *
 - o The large set of terminologies used for energy communities lead to “confusion and makes horizontally comparing both policy recommendations and research findings cumbersome and less stringent” (p.7)
 - o A clear definition and requirements for those concepts would improve policy analysis, recommendations and translation of lessons learnt (p.7)
- Interest in ECs is increasing, but the public awareness of the concept is low (p.7)
- “[...] the studied countries have not found a conclusive way to circumvent the conflict between economic viability for the communities and fairness in shared costs for the collective grid” (p.8)
- All studied countries have existing legislation that regulate the initiation, activities and organisation of ECs (p. 26)
- The electricity act is the main legislative framework in Sweden- regulates the energy market and the implementation of energy communities (p.26)
 - o *Energimyndigheten* is responsible for promoting energy-efficient measures and investments in RE technologies (p.26)
- Four different EC models in Sweden (p.29):
 - o Wind power cooperatives
 - o Eco-villages
 - o Solar power cooperatives (within *bostadsrättsföreningar*)
 - o Rural communities run small scale heating systems
- General and country/model specific **barriers**:
 - o Lack of awareness and knowledge among local, regional, and national policy makers and the general public (p.30)

- Prevents municipalities from assigning space for energy communities or allowing communities when they take initiative
 - Creates barriers in accepting new ECs or finding members for ECs
 - Mitigation: A representative body that facilitates the knowledge transfer between ECs and policy makers (p.34)
 - Lack of governance framework for energy communities (p.33)
 - Mitigation: A representative body of ECs to lobby for the inclusion of energy communities in the governance framework (example: Rescoop at EU level) (p.33)
 - Lack of technical or legal knowledge required amongst community members (p.30)
 - Large upfront investments – especially for low-income households (p.31)
 - Mitigation: Support from governments in the form of funding or subsidies (p.34)
 - The Clean Energy for all Europeans package gives ECs the right for energy sharing (p.31), BUT:
 - Energy sharing is however not defined and is up to the member state to decide (p.31)
 - In Sweden, energy sharing is only possible when a microgrid has been deployed in a building by a company or a district system operator (p.32)
 - There is an exemption in the law in Sweden for building microgrids that only applies to small internal networks in an easily defined area. A few pilot projects have been permitted to develop a common energy system for residential areas (p.32 & p.68) – legal barrier for EC development
 - Energy prices in Sweden are evenly distributed among those connected to the grid to avoid higher costs for maintenance of the grid in low-density parts of the country – If allowing the creation of micro-grids, the network operators would lose a part of their income (p.32)
 - Technological barriers related to lack of digital infrastructure + slow rollout of smart meters (p.33)
- General and country/model specific **enablers**:
 - Some governance frameworks can actually enable ECs (p.35)
 - Possibilities for ECs to reduce the energy load of local grids (p.35)
- **Drivers** for founders and members of ECs:
 - The control over their own energy supply (p.35)
 - The prospect of relatively stable and low energy prices (p.35)
 - The prospect of security of supply in rural areas (p.35)
 - The chance to contribute to the energy transition (p.36)
 - Democratisation of and benefits for the community (p.36)
- When a country already has a high share of renewable energy in its energy mix, citizens are less likely to engage in the establishment of energy communities (p.36)
- Country study Sweden (pp.65-74)
 - No current examples of ECs adhering to the definitions of CEC or REC (p.65)
 - The Swedish Energy Market Inspectorate is responsible for the implementation of the EU directives (p.67)
 - Barriers:
 - DSO's monopolistic position on the distribution market is a practical barrier (p.69)
 - Various actors' economic and technical constraints (p.69)

- Generally low level of public knowledge (p.69) regarding energy consumption and unawareness of solutions to lower costs (p.70)
- Political and judicial ambiguity related to the definition of ECs (p.69)
- General lack of interest from DSOs to engage in projects aiming at developing energy communities
- Interviewees of the study expect the national legislation soon to be adapted, to enable ECs to produce, share and store energy (p.69)

Sweden Europe barriers drivers

1.3. MAGNUSSON AND PALM, 2019

Magnusson, D. and Palm, J., 2019. Come together – The development of Swedish energy communities. *sustainability*, [e-journal] 11(1056). <https://doi:10.3390/su11041056>.

Summary: This article analyzes the existing community energy projects and discusses the potential community energies have in the Swedish energy transition perspective by using interviews, document studies, analysis of previous studies as well as website analysis. The results show that fewer initiatives have been taken in Sweden than in other countries, but that even with a rather 'hostile' institutional setting CE has emerged as a phenomenon.

- Sweden's institutional setting is not as beneficial for CE development as in other countries, despite the already high share of RES in the energy mix which is not always the case in other countries with a higher share of CEs (p.2).
 - Fewer CE initiatives in Sweden than in many other countries (p.15)
 - Unfavorable conditions for starting CE initiatives with the existing structure of the energy market and the high involvement of municipalities (p.15)
- Most studies focus in success stories of CEs, but failed projects could help to understand barriers and institutional limitations (p.4)
- "Energy policies and economic support are among the strongest factors that influence the success rates for CEs." (p.12).
 - Specific investment support systems have created a window of opportunity for entering the market with renewable energy, especially true for wind cooperatives (p.12).
- **Drivers:**
 - The interest in taking things into their own hands, hence the access to RES through own production (p.9)
 - Big drivers in especially rural communities are environmental concerns and many have been inspired by the anti-nuclear, biological cycle or climate change movement (p.10).
 - The strongest aims and goals that were identified focus on the economical aspect, like the collective production of green energy, energy and money savings, rather than spreading knowledge or self-sufficiency (p. 10).

- Competence within the community to navigate through bureaucratic matters and standards as well as the planning process (p.11)
- Good networking skills is another success factor, to attract new members or get support from e.g., energy companies, which is especially important for solar PV cooperatives (p.11)
- Legislation and long-term rules for investment in RE
- **Barriers:**
 - The navigation through bureaucracy, grants applications, technical system configurations and standards (p.11)
 - The Swedish state has made it more difficult to run a cooperative, due to new regulations, which has affected the interest to start cooperatives (p.12).
 - The low electricity prices as well as the decreased price for certificates of the green electricity certificate system has led several cooperatives to stop (pp.12f.).
 - When plants have reached the end of their lifespan, CEs have stopped the project due to economic insecurities (p.13).
 - Most of the subsidies and tax reliefs in RE investments focus on either private costumers or corporations, but not CE initiatives (p.13).
- A special aspect for CEs in Sweden compared to other countries is that sometimes membership in initiatives have not been restricted geographically (p.15).

rural

urban

Sweden

barriers

drivers

1.4. PALM, 2021

Note: The following article/report is worth looking into. Good summary of ECs and the barriers and drivers of their emergence in the named countries (including Sweden) and earlier research on ECs.

Palm, J., 2021. Energy communities in different national settings – barriers, enablers and best practices. [pdf] Newcomers. Available at: <[https://www.newcomersh2020.eu/upload/files/Deliverable%203_3_%20Energy%20communities%20in%20different%20national%20settings_barriers%2C%20enablers%20and%20best%20practices.p](https://www.newcomersh2020.eu/upload/files/Deliverable%203_3_%20Energy%20communities%20in%20different%20national%20settings_barriers%2C%20enablers%20and%20best%20practices.pdf)
df> [Accessed 4 September 2023].

Summary: Aims to explore how new clean energy communities (ECs) develop, under which polycentric settings ECs evolve and under what conditions such initiatives are suppressed. At a national level, the project will assess socio-economic, technical, institutional characteristics and actors supporting the emergence of new ECs, by describing the national characteristics of the countries in focus and making use of earlier research. The focus lies on the following member countries of Newcomers: United Kingdom, Germany, Netherlands, Sweden, Slovenia and Italy.

- Newcomers' definition of energy communities: "an association of actors engaged in energy system transformation for reduced environmental impact, through collective, participatory, and engaging processes and seeking collective outcomes." (van der Grijp et al., 2019, p. 23) (p.15)
- ECs are more common in Northern Europe compared to southern Europe (p.15)

- The included countries – UK, Germany, Netherlands, Sweden, Italy and Slovenia – often lack a national definition of EC's which is why the exact number of existing ECs can differ in different studies (p.16)
- Sweden does not have a legal framework in place for ECs (p.16)
- Rural areas have more space, where wind as an electricity source is more suitable. Urban areas are more dense and solar PVs are more suitable here (p.18)
- **Barriers:**
 - "Another deficit is that a lack of one unanimously agreed definition contributes to an ambiguous usage of the concept, leading to confusion and overlooks differences in the models that can be important for understanding how ECs emerge" (p.16)
 - The lack of technical knowledge has been detected as a barrier for participating in ECs (p.19)
 - The lack of expert knowledge will also continue to be a barrier, with technology becoming more complex (p.19)
 - The lack of trust in others and to the legal and political system (p.21)
 - Financial resources of single households to start participating in an EC project (p.22)
 - People already owning a PV installation are less willing to participate in an EC (p.24)
 - Access to cheap domestic energy sources poses a barrier for ECs (p.24)
 - A legal framework that allows ECs to compete "on an equal footing", even though RED II obliges the member states to create one (p.29)
 - Financing ECs and their projects is one of the biggest barriers, especially in the development and planning stage (p.30)
 - ECs need access to subsidies, bank finance, grants (p.31)
 - Lack of political support and lack access to politicians and policymakers (p.34)
 - Political discourse became a barrier for ECs in Sweden and Netherlands, whereas in Denmark it became an enabler (p.34)
 - The structure of the electricity market is important for the emergence of ECs. Centralized design and regulation of existing energy systems is a main barrier for EC emergence – monopolized grid or no grid access (p.35)
- **Drivers:**
 - Grassroot innovations emerge from existing experiences and knowledge within the community rather than through education and research (p.19)
 - High GDP/capita which makes it possible for households to join an EC (p.21)
 - The existence of microgrids for peer-to-peer markets has been identified as an enabler for EC to emerge (p.28)
 - A liberalized electricity market with domestic competition has been identified to be more beneficial for ECs (p.36)
 - Support of an umbrella organization to help put ECs on the political agenda, like it is the case in Germany, Netherlands and UK, but not in Sweden (p.37)
- **Opportunities:**
 - ECs can contribute to its members saving money and energy and could bring money to low-income households as well as contribute to the collective distribution of benefits (p.21)
 - Virtual net metering might open up new innovative solutions for ECs, if legal regulations allow for it (p.27)
 - Virtual power plants are an established business model which is getting more attention, by established ECs, who are looking into combining existing renewable generation projects within their local area (p.27)

- ECs have the potential to change the whole owner-structure of the electricity market (p.36)
- No obvious patterns or correlations between the number of transmission system operator or distribution system operator and the number of existing ECS in a country (p.26)
- A FiT scheme has both been a barrier and a driver in EC development. Sweden never had a FiT scheme. Other countries cancelled the FiT scheme after a couple of years (pp.31ff.)

Sweden Europe barriers drivers

1.5. NORTHERN PERIPHERY AND ARCTIC PROGRAMME, N.D.

Northern Periphery and Arctic Programme, n.d. *PESTLE Analysis of Barriers to Community Energy Development*. Northern Periphery and Arctic Programme. Available at: <https://leco.interreg-npa.eu/subsites/leco/PESTLE_Analysis_LECO_A4_190110-singlepages.pdf> (Accessed 6 September 2023).

Summary: A PESTLE analysis, which is used to analyze and monitor the macro-environmental factors that have an impact on an organization or sector. This analysis has identified common barriers to community energy projects specific to the member regions of the project, including northern Sweden.

Barriers to community energy (Copied and pasted p.5):

- **Societal, cultural, political and/or organizational:**
 - Lack of historic experience with cooperatives and civic activism
 - Low trust in the cooperative model as a viable alternative
 - Lack of political support from local representatives
 - No experience with setting up cooperatives
 - Organizational challenges – pre-planning stage barriers
- **Legal, administrative, bureaucratic:**
 - Complicated legal framework, high levels of bureaucracy to acquire licenses
 - Lack of national community energy strategy;
 - lack of national targets for community energy projects, which then are broken down in Local Energy Action Plans by local authorities
 - Bureaucratic barriers to grid connection (complicated application procedures, uncertainty of approval, costs, time consuming)
 - Not allowed to operate micro-grids - producing, own-use, selling within community, selling to third-parties – as compared to only: sell it to the grid and buy it back (often with low financial returns to the community – profits are again made by companies outside the community, which defeats the idea to keep revenue within the community)
 - Lack of supportive local authorities and/or local energy agencies
 - Generally, no support schemes for Renewable Energy Sources (RES) projects
- **Technical:**
 - Technical challenges – lack of expert knowledge to design, plan, procure, implement, commission a project
 - Lack of expert knowledge for operation and maintenance

- Size of energy project
- **Financial:**
 - Financial challenges in the initial stages of project development; access to finance, grants, etc.
 - Fair and secure payments for energy generated (insufficient Feed-in-tariffs (FIT), FIT only for wind, but not for Solar PV, no standardized PPAs, third-party-offtake not possible)
 - Insufficient incentives for renewable heat projects: replacing fossil fuel heating with biomass boilers or solar thermal, heat pumps
 - Complicated tax rules, no tax exemptions
 - Generally, no tax incentives for RES projects, lack of guarantees Challenges in mature cooperatives:
 - Expansion of power generation, of number of members – how shall older and new membership shares be valued?
 - Re-investment into existing installations
- **Barriers specific to rural areas in northern Sweden (p.22):**

SWEDEN: Barriers to Community Energy Projects	
POLITICAL BARRIERS: • A small number of stakeholders who needs to be involved in many processes – problem in terms of time, but also of only a few persons representing a whole county. • Lack of effective funding mechanism for environmental investments. • Lack of regional (legal) instruments for pushing the shift towards low-carbon communities. • Focus on short-term solutions instead of long-term structures.	ECONOMIC BARRIERS: • Energikontor Norr as regional energy agency mainly project financed, what makes long-term planning of service offers difficult. • Lack of funding instruments for investments in energy and climate projects, e.g. passive houses. • Not enough governmental support for e.g. investments in public transport infrastructure.
SOCIAL BARRIERS: • Lack of capacities in small municipalities: staff, knowledge and money. • Personal meetings contributing to build up confidence and ensuring continuously collaboration are difficult due to long distances. • More people from the rural areas moving to cities and the coast, thereby weakening small municipalities even more. • Norrbotten's countryside is characterized through traditional thinking what means that new and innovative services are difficult to implement and push. • Failure in building up working multi-level governance structures lead to less engagement from the few people engaged in energy and climate work. • Interest in energy, environment and climate issues may decrease.	TECHNOLOGICAL BARRIERS: • High energy demand of industry (mining, steel, paper mills) and not enough people for public transport in remote regions.
LEGAL BARRIERS: • Divided competencies between two regional authorities ("Region Norrbotten" and "Länsstyrelsen Norrbotten").	ENVIRONMENTAL BARRIERS: • Long distances and sparsely population make collaboration difficult, time- and cost intensive. • Conflict with other targets, e.g. nature protection and economic growths.

Sweden Europe barriers rural

1.6. CAMPOS, ET AL., 2020

Campos, I., Pontes Luz, G., Marín-González, E., Gähns, S., Hall, S. and Holstenkamp, L., 2020. Regulatory challenges and opportunities for collective renewable energy prosumers in the EU. *Elsevier*, [e-journal], 138(2020). <https://doi.org/10.1016/j.enpol.2019.111212>.

Summary: This study carries out a cross-country comparison between the regulatory frameworks of nine European countries and regions -Belgium, Croatia, France, Germany, Italy, Portugal, Spain, Netherlands and the UK – to reveal the main challenges and opportunities that these have posed to collective RE prosumers. The result show that four countries have a more favorable framework (France, Germany, Netherlands, UK), but also that the current legal framework at the EU level does represent a clear opportunity for collective prosumers.

- **Regulatory challenges:**

- The lack of ability to legally set up a renewable energy community (p.8)
- The lack of incentives to set up jointly acting renewable self-consumer projects (p.8)
- In some cases, the reduction or removal of existing incentives, like feed-in-tariffs (p.8)
- Some countries do not allow collective self-consumption schemes (Belgium, Croatia, Italy) (p.8)
- “Also, collective self-consumption laws tend to have a one size fits all approach and set spatial limits and/or restrictions on installed capacity that can inhibit the innovation potential of RECs.” (p.9)
- Permitting processes are too complicated
- Frameworks not flexible enough to create new solutions

- **Opportunities:**

- Flexible regulatory frameworks would make it possible for communities to find creative ways of taking advantage of untapped opportunities (p.9).
- Virtual net metering could potentially open up a range of opportunities for collectives (p.9)
- “Across Europe, groups of citizens are developing new modes of governance, business models and alternative financial mechanisms to enable the production and self-consumption of renewables, even when legal frameworks pose barriers” (p.9)

- **Recommendations:**

- A specific legal framework to address the complexity of RECs and their needs (p.9)
- New laws that include and allow for innovation and experimentation (p.9)
- Prevention of the exclusion of vulnerable communities and low-income families in new REC projects (p.9)

Europe barriers

1.7. COY, ET AL., 2021

Note: I think it is beneficial to read through the drivers of empowerment (p.6-8). Bullet points will do no good here.

Coy, D., Malekpour, S., Saeri, A.K. and Dargaville, R., 2021. Rethinking community empowerment in the energy transformation: A critical review of the definitions, drivers and outcomes. Elsevier. [e-journal], 72(2021). <https://doi.org/10.1016/j.erss.2020.101871>.

Summary: This paper synthesises the literature on community empowerment for sustainable transformations and explores how community empowerment can be operationalised in the energy transformation.

- Definition of empowerment connected to sustainable transformation according to the article (outcome from the research): “the process of an individual or group or community increasing their capacity and contextual power to meet their own goals, leading to their transformative action” (p.10)
 - Two key components of shifting political power: building the individual or groups’ capacity and changing the contextual power (p.10)
- **Barriers:**
 - Power plays a crucial role in shaping how the transformation unfolds and who can participate, and it is often the voices of those in power (the elite actors) that are heard, rather than the communities (p.2). Power holders prevent communities from accessing resources or challenging norms (p.8)
 - There needs to be broader financial and structural changes in order for communities to change their energy engagement to be less carbon intensive (p.2)
 - Community energy groups are often unable to reshape the existing political power structures, such as the incumbent centralised energy actors, business models and regulatory frameworks (p.2)
 - Poor living conditions as a barrier for communities to become empowered (p.9)

- **Drivers of empowerment** on different levels (individual, interpersonal, community, societal and policy and systems):

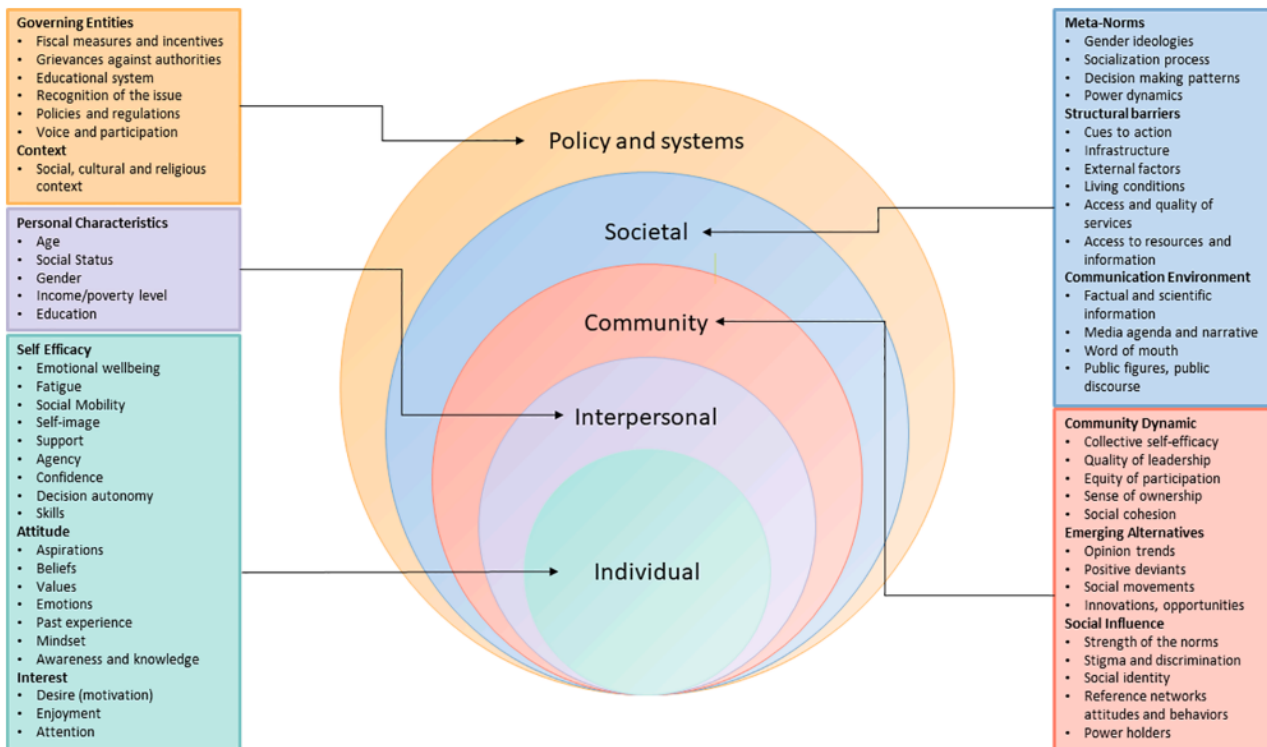


Fig. 7. Drivers of empowerment, categorised by their level of action, boxes contain the specific drivers.

- “Meta norms are the underlying ideologies deeply entrenched in culture, identity, influencing all other drivers. For empowerment to occur, established meta-norms often had to undergo and evolution wherein traditional power dynamics, decision-making patterns and gender ideologies were subverted.” (p.8)
- The control and access to resources made communities gain power and contributed to transforming decision-making patterns (p.8)
- Outcomes of empowerment in the following steps: Participation > Agency > Autonomy > Power-Shift
 - Participation: “the action of taking part in something” (p.9)
 - A key indicator of empowerment is agency: “the capacity of individuals to act independently and make their own free choices” (p.9)
 - Autonomy: “the right of a group to govern themselves or organise their own activities” (p.9)
 - Power-shift: The community is in total control and has power over resources and to innovate; they are the owners of the problem (p.9); the highest form of empowerment (p.11)
- “facilitating community empowerment within the energy transformation requires a transdisciplinary approach building on the commonly recognized drivers of engaging but also focusing equity of participation.” (p.10)
- Linking the concept of empowerment to energy engagement and citizenship provides a mechanism for ensuring inclusivity (p.11)

1.8. KOUIJ, ET AL., 2018

Kooij, H.-J., Oteman, M., Veenman, S., Sperling, K., Magnusson, D., Palm, J. and Hvelplund, F., 2018. Between grassroots and treetops: Community power and institutional dependence in the renewable energy sector in Denmark, Sweden and the Netherlands. *Elsevier*, [e-journal] 37(2018) pp.52-64. <http://dx.doi.org/10.1016/j.erss.2017.09.019>.

Summary: This paper tries to understand the differences in the emergence of GIs for renewable energy in relation to the institutional characteristics of Denmark, the Netherlands and Sweden. The possibilities for GIs to emerge were analysed in three dimensions: the material-economic, the actor-institutional and discursive dimension. The authors used policy documents, research reports, available data and interviews.

- “Grassroot initiatives (GIs) include local and regional initiatives, established by groups and communities that aim for a more durable energy system in their direct environment” (p.1)
- Sweden
 - Barrier:
 - Sweden creates an environment where the local government and municipalities are very active in providing renewable energy. Since this would be the task for GIs, there is not always the need for GIs to develop (p.2)
 - Electricity system consists mainly of hydro and nuclear (p.6)
 - Regulations and economic incentives have restricted the possibilities for small actors to enter the electricity market (p.7)
 - GIs were limited due to monopolies before 1996. Strong municipalities in both electricity and heating sector and low energy prices (p.7)
 - “[...] when renewable energy is completely competitive, it will be adopted into the (large-scale) central energy production structure and there is no space or incentive for GIs, and no option to enter the energy market. This is the case in Sweden: the affordability of hydropower combined with the large investments required created a clean, market oriented and centralized energy provision system in which nuclear power is supplemented by hydropower.” (p.8)
 - “In Sweden, the political culture is dominated by the strong welfare state and the longstanding hegemony of the Social Democratic Party, which implemented sustainability programs in a topdown and bureaucratic manner” (p.9)
 - “In a ‘passive’ understanding of citizenship, the responsibility for sustainability issues is placed with the state.” (p.10)
 - Wind cooperatives are struggling due to low electricity prices and sometimes they are not eligible for green electricity certificates any longer, as the plants are older than 15 years – no cooperatives created in recent years and lower incomes from production (p.11)
 - Driver:
 - Eco-villages were created during the anti-nuclear movement (p.7)
 - A reason for people in Sweden to become a prosumer has been to resist a central electricity system, governed by a few multi-national companies (p.7)

- A change in legislation in 1997 made it possible to sell own-produced electricity back to the grid, increasing renewable electricity production by prosumers through wind and solar PVs (p.7)
 - The green electricity certificate system was implemented in 2003 and supported the development of RE production (p.7)
- Denmark:
 - GIs were started around the 1973 oil crisis to introduce the idea of RES to challenge the dominant energy discourse and nuclear plans (p.3)
 - Knowledge exchange between GIs and government led to energy plans without nuclear in the picture (p.4)
 - From 1986-2000, local ownership was supported by institutional requirements, leading to an increase of privately owned shares (pp.4f.)
 - The large increase of wind power share in domestic electricity consumption led to market losses for energy companies. Now they are trying to lobby against local co-operative ownership of more wind power (p.5)
 - A new government from 2002 onwards, started a policy against green energy technologies, resulting in removed funding and nearly no new wind turbine installations between 2003 and 2007 (p.5)
- Netherlands:
 - Mainly fossil energy system with heavy reliance on domestic natural gas (p.5)
 - RES not seen as a way to replace fossil fuels in the Dutch energy mix in 1980, which motivated citizens to start the umbrella organization ODE that started a network for citizens that built their own wind turbines (p.5)
 - GIs emerged for the simple reason to generate profits for the local community (p.5)
 - Another push for GIs happened after the liberalization and after the increased awareness about climate change (p.6)
 - GIs develop in the absence of governmental support (p.6)
- Overall barriers and drivers:

Table 2
Conditions of possibility for GIs in the material-economic dimension.

	Constraining conditions for GIs	Enabling conditions for GIs
Structure of economy	Industrialized economy with centralized energy production and tax exemptions for energy intensive industry	SME economy, decentrally organized energy infrastructure and provision
Energy dependency	Availability of cheap domestic (fossil) central energy sources, fossil fuels state revenues	High dependence on (fossil) fuel imports: international dependency
Renewable energy potential	Low (small-scale) RE potential due to biophysical conditions	High RE potential due to biophysical conditions
Costs of renewable energy	High (installation) costs of RE compared to conventional energy	Low (installation) costs of RE compared to conventional energy
Energy prices	Energy prices set by market mechanisms, no correction for externalities. Or: competitive RE prices enabling market parties to engage in RE	Feed-in tariffs, subsidies on RE, CO ₂ taxation

Table 3
Conditions of possibility for GIs in the actor-institutional dimension.

	Constraining conditions for GIs	Enabling conditions for GIs
Market orientation	International market competition and lobby aimed at international competitiveness (industry policy)	National market orientation
Market rules and resources	Closed energy market; rules and resources tailored to large installations and players	Liberalization of energy market, increased domestic competition
Local ownership rules	No rules for supporting/protecting local and consumer ownership, or very restrictive rules	Rules supporting/protecting local and consumer ownership
Grid access and remuneration rules	Monopolized grid, no/uneconomic grid access, no guaranteed minimum prices	Affordable grid access or cooperating energy companies, guaranteed prices
Energy companies	Large energy companies that dominate energy production and distribution	Small energy companies (including consumer owned), 'Green' signature energy companies
Ownership energy companies	State owned energy companies	Consumer-owned energy companies
Business sector	Business sector aggressively developing RE projects	Competition and unbundling
Political culture and decision making	Closed political culture, no access to (influence) decision making	Minor share of businesses engaged in RE technology development early on
		Strong networks between GIs, knowledge production and sharing, policy and political influence and support

Table 4
Conditions of possibility for GIs in the discursive dimension.

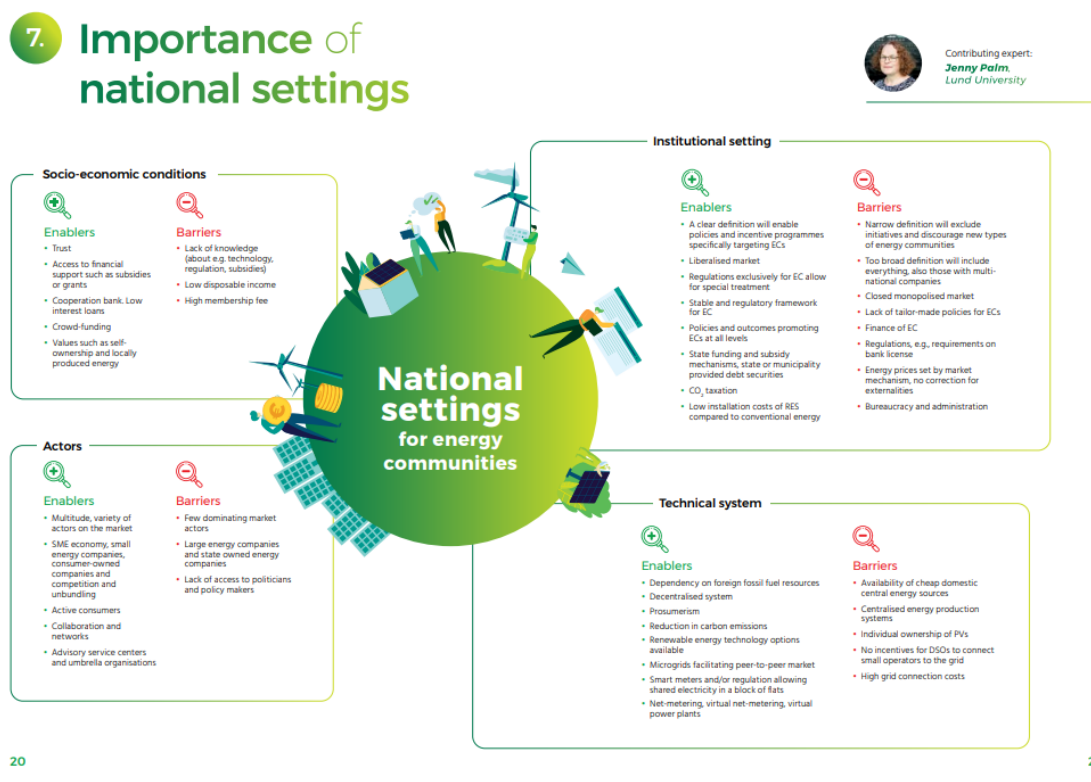
	Constraining conditions for GIs	Enabling conditions for GIs
Dominant discourse	Economic: affordable energy and security of supply	Sustainability: climate change mitigation and environmental protection
Openness to non-dominant discourses	Non-dominant issues such as sustainability and decentralization are not politically recognized	Non-dominant issues are acknowledged
Accessibility to political arena	Alternative discourses are not voiced in political arena or not acted upon	Alternative discourses are seen as feasible arguments. Or: complete lack of acknowledgement pushes citizens into protests
Knowledge creation and distribution	Monopoly of knowledge production of energy companies and research institutes	Democratic distribution of knowledge production
Perceived feasibility of technological innovations	Little importance attributed to alternative technologies and innovations	Positive image of innovation of technologies and collaborations between actors
Vision of energy future	Nuclear <i>Leitbild</i> , vision of centralized energy production	Open to alternative resources and scales of energy production
Culture and perspective on citizenship	Individualistic culture, 'passive' citizenship norms: state is responsible, citizens as consumers.	Cooperative tradition/culture, 'active' citizenship norms, individual is responsible, citizens as prosumers and owners of RE facilities.

Sweden Europe barriers drivers

1.9. DREVENŠEK AND TAJNŠEK, N.D.

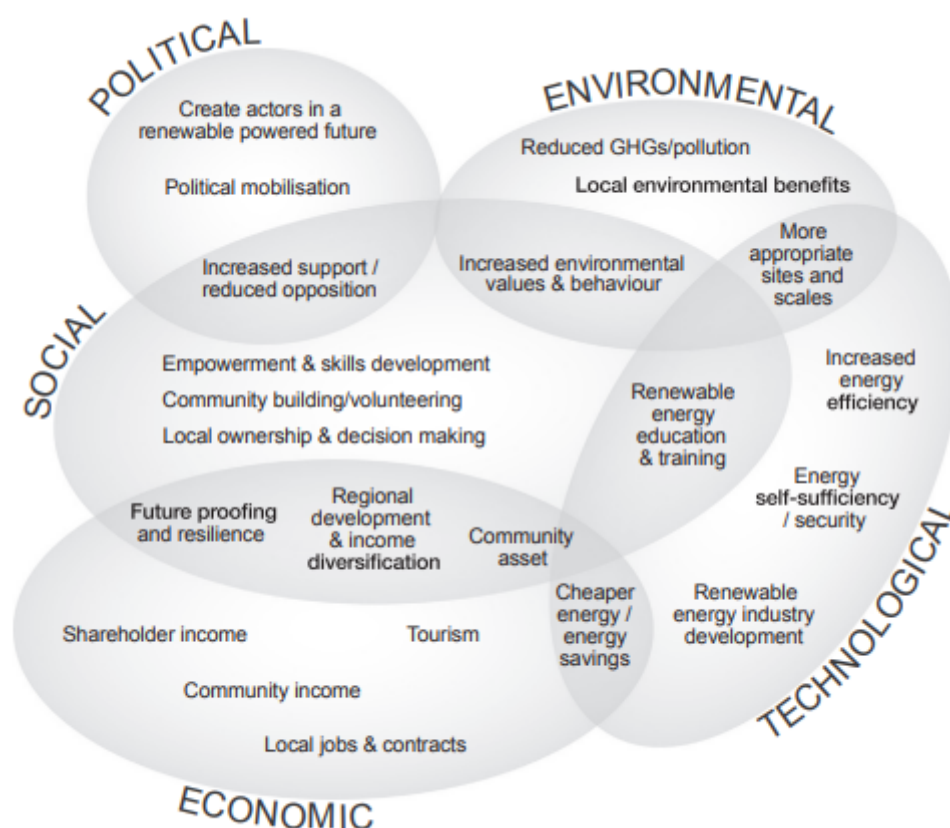
Drevenšek, M. and Tajnšek, I., n.d. *Handbook – Energy Literacy for Energy Communities*. [pdf] Newcomers. Available at: <[NewcomersHandbook_spread_fin.pdf \(newcomersh2020.eu\)](#)> [Accessed 29 September 2023].

- Jenny Palm listed enablers and barriers the importance of national settings for energy communities – Was mentioned in other articles as well, but is a good overview:



Hicks, J. and Ison, N., 2018. An exploration of the boundaries of “community” in community renewable energy projects: Navigating between motivations and context. Elsevier, [e-journal] 113(2018) pp.523-534. <https://doi.org/10.1016/j.enpol.2017.10.031>.

- Diverse forms of community renewable energy (CRE) (p.1)
- A common understanding of CRE is needed (p.1)
 - o Lack of clarity is a risk factor (p.2)
- A broad definition could provide CRE actors the adaptability needed to develop in contextually appropriate ways but also leave openings for frauds to take advantage of the community brand and tarnish the concept of CRE for other communities (p.2)
- Community energy: “Projects where communities (of place or interest) exhibit a high degree of ownership and control of the energy project, as well as benefiting collectively from the outcomes (either energy-saving or revenue-generation).” (p.2; cited from Seyfang et al., 2013)
- Community renewable energy: A more specific subcategory of CE (p.2)



- Contextual factors are critical to shaping the final form and outcomes of a CRE project (p.5):
 - o Physical factors: topography, renewable resource availability and existing infrastructure, resource availability

- Technology factors: Cost of RE technologies, the maturity and modularity of the technology, the energy needs and demand profile of the community/building, scale opportunity
- Institutional factors: Structure of the energy market, regulatory environment, laws governing legal structures, renewable energy policies (especially incentives for RE and CRE), culture within existing energy and other relevant institutions
- Community factors: Local history and culture (in a CRE project based on community of locality), relationships or social capital, and skills and knowledge available. Social perceptions of and appetite for certain technologies.
- “Motivations are the foundation from which CRE projects are brought into existence” (p.5)
 - The motivation can be different depending on the context (anti-nuclear movement in Denmark and climate change in Australia)
- CREs and their structure can look very differently (pp.6ff.):
 - from only local members to organizations and businesses
 - from one vote per actor to one actor having all the votes
 - from the financial benefit being used for the community to investors gaining surplus income
 - from the scale of the technology being in relation to local needs to maximizing the economic outcome
 - from local engagement starting early in the process to very little engagement
 - all of the above can be seen having the “community status” (p.8)
- differentiating between stronger and weaker forms of CRE is crucial for more targeted and conscious applications of CRE in policy and project contexts

Europe

drivers

1.11. MIGNON AND RÜDINGER, 2016

Mignon, I. and Rüdinger, A., 2016. The impact of systematic factors on the deployment of cooperative projects within renewable electricity production – An international comparison. Elsevier, [e-journal] 65(2016) pp.478-488. <http://dx.doi.org/10.1016/j.rser.2016.07.026>.

Summary: A comparison of the deployment of RE cooperative projects in France, Germany and Sweden, done by a review of literature and qualitative interviews.

- The concept is still quite unclear, but the following understanding is shared: “RE cooperatives, no matter the context, imply some involvement of the public (e.g. the local community or the citizens of a region or country) in the decision-making (e.g. through democratic governance or through the ownership of the majority share of the projects) and a distribution of the benefits generated by the RE project in question.” (p.2)
- Opportunities:
 - Facilitation of energy transition by creating local acceptance of wind power, by raising equity for the development of RE plants and by creating awareness of energy issues (p.2)
 - Lowering obstacles associated with the development of RE projects (p.2)

- When initiative is taken to develop RE projects, it is shown that the local and public acceptance of RE increases (p.2)
- Creation of an additional equity source as participants in RE cooperatives are less affected by risk fluctuating profits and unstable policies (p.2)
- Some authors believe that RE cooperatives contribute to the democratization of the energy system through decentralization of energy utilities (p.3)
- Barriers mentioned in previous studies (p.3):
 - Conflicts among members or within the community
 - Lack of professionalism
 - Lack of legitimacy
 - Lack of access to locations,
 - Unstable policy contexts
 - Lack of experience
 - Lack of access to capital
 - Lack of networks
 - Etc.
- The development and diffusion of innovations is determined by factor, e.g. strengths and weaknesses at the socio-technical level of the system – Negro et al. (2012) proposed the theoretical typology of systematic factors with an impact on the deployment of RE technologies (p.3):
 - Market structure
 - Infrastructures
 - Hard and soft institutions
 - Interactions between actors
 - Capability factors
- Barriers mentioned in this study:
 - Lack of transparency regarding distribution costs in cases when grid reinforcements are needed – A new RE project has to carry the costs of its connection, no matter the voltage level or potential social benefits – One project carries all the costs (p.5)
 - Dominant utilities are advocating toward policies in favour of their own business model or large market actors (p.6)
 - The money raised in order to finance RE cooperative projects comes exclusively from selling shares to individual members, because banks have been reluctant to grant loans (p.5)
 - the amount of shares that can be bought by members has to match their electricity consumption (p.8)
 - RE projects often meet local opposition, even though it has general support in public opinion (p.5)
 - The tradeable green certificate is favourable toward RE development in general, it has received criticism for its inability to drive technological innovation (only the most mature technologies are awarded support) (p.6)
 - Lack of transparency of permit procedures of the policy system, particularly for wind power (p.6)
 - Prosumers still have to pay the energy tax for the electricity they consume (p.6)
 - Systemized knowledge opportunities are rare (p.6)
- Drivers:
 - Traditional actors in the Swedish energy market have worked in favour of decentralized projects (p.5)

- Grid connection process considered relatively simple, compared to other countries (p.5)
- RE cooperatives can rely on two complementary sources for debt capital in Germany (p.5)

Sweden Europe barriers drivers

1.12. KLEIN AND COFFEY, 2016

Note: Very theoretical study. For me difficult to understand.

Klein, S.J.W. and Coffey, S., 2016. Building a sustainable energy future, one community at a time. Elsevier, [e-journal] 60(2016) pp.867-880. <http://dx.doi.org/10.1016/j.rser.2016.01.129>.

Summary: A review of the theoretical basis for community energy and a review of community energy literature.

- Individual energy consumers are more likely to adopt energy behaviours and technologies that have been adopted by people in their own social network with similar beliefs, attitudes, education and status (p.3)
- Grassroot innovation: “The social innovation of creating a network of activists to lead bottom-up sustainable development solutions that respond to local situations and community interests” (p.3)
 - CE “initiatives comprise a niche of sociotechnical innovation in which people practice new social infrastructure, institutions, value sets and priorities in a value set that is distinct from the dominant regime” (p.3)
- CE initiatives could act as a building block for broader societal changes toward sustainable development
- “In this way, Social Practice Theory supports the development of community energy as a means to change individual behavior because engaging individuals in a grassroots initiative disrupts the “normal way of life”, causing individuals to practice the preferred behavior without expecting a prior individual shift in values. This social engagement may in turn lead to a shift in individual values after the behavior has been practiced.” (p.3)
- Groups often perform better than the sum of their parts (p.3)
- “[...] energy behavior change requires alignments of multiple processes at multiple levels of sociotechnical interaction and that socio-technical innovations at the niche level may encourage diffusion of innovation at the regime level” (p.3) – multi-level perspective
 - 3 key processes for successful niche growth: managing expectations, building social networks, and learning)
 - Expectations: widely shared, specific, realistic and achievable.
 - Networking activities: include many different stakeholders, who share information and/or financial resources.
 - Learning: include evidence of “second-order learning” (questioning regime assumptions and constraints)

- 3 ways in which niches can influence the regime: replication, growth in scale, and translation
 - Replication: individual projects contribute to aggregative changes.
 - Scaling-up: individual projects grow in scale and attract more participants from beginning to end of project timeline.
 - Translation: niche ideas become integrated into mainstream settings
- CE definition: many definitions, depending on the context (p.4)
 - “a project or program initiated by a group of people united by a common local geographic location (town level or smaller) and/or set of common interests; in which some or all of the benefits and costs of the initiative are applied to this same group of people; and which incorporates a distributed energy generation technology (for electricity, heat, or transportation) based on renewable energy resources (solar, wind, water, biomass, geothermal) and/or energy conservation/efficiency methods/technologies”
- Key insights from studies about CE:
 - Transition town movement: “mobilize community action and foster public empowerment and engagement around climate change” (p.5) – TT movement is an example of successful niche formation in several respects, but there are significant weaknesses in the niche processes, which limit the potential for this grassroots innovation to achieve more widespread diffusion (p.6)
 - CE provides a useful way for citizens to engage in grassroots action, there is a need for more development or organizational structures and goals in order to achieve lasting results (p.6)
 - Gain (increase/protect resources) and normative (comply with social norms, ethics and morals) considerations drive participation in community energy initiatives, including decreasing energy costs and reducing climate change threats (p.6)
 - Trust is an important condition for the success of local energy projects (p.6)
 - Some initiatives have been successful when they link their initiatives to shared community assets, such as schools, which become a source of community pride and an educational opportunity (p.6)
 - Barriers of wind projects in USA: collapse of financial sector, difficulty in securing construction financing, public opposition due to scenic or cultural property and endangered species
 - Opportunities of wind projects in USA: federal investment tax credit, construction loans, being surrounded by other large wind projects, having examples of similar projects, government support in securing transmission lines, guidance from authorities, and investment by private individuals and companies

1.13. FRIENDS OF THE EARTH EUROPE, RESCOOP.EU AND ENERGY CITIES, 2020

Friends of the Earth Europe, REScoop.eu and Energy Cities, 2020. *Community Energy - A practical guide to reclaiming power*. [pdf]. Available at: <[Community Energy: A practical guide to reclaiming power - REScoop](#)> [Accessed 26 September 2023].

Summary: A guide for developing community energy in Sweden.

- CE is unequally developed in Europe (p.132)
- Projects need good laws and rules to support them – CE projects flourish better in those countries where it is easier to set them up (p.132)
- Barriers:
 - o Laws often change (p.132)
 - o Internal conflicts in the project group (p.134)
 - o Lack of finance (p.134)
 - o Getting planning permission (in some cases) (p.134)
 - o Administrative burden – time consuming (p.135)
 - o Grid connection (p.135)
 - o Lack of understanding what CE is – in some countries CE is a very new concept (p.136)
 - o Local opposition to renewables (NIMBY) (p.136)
 - o NOTE: the guide makes suggestions on how to overcome all of these barriers or even prevent them from occurring
- Drivers:
 - o CE is recognised in EU law since 2018 – it is a countries responsibility to ensure that unfair barriers do not block the path of CE projects (p.132)
 - o National governments must create an enabling framework to support community energy in their country (p.133)
 - o There must be one place for citizens to go for advice, to gain permits to participate in the energy system (p.133)

1.14. ROBERTS, ET AL., 2014

Roberts, J, Bodman, F and Rybski, R., 2014. Community Power: Model Legal Frameworks for Citizen-owned Renewable Energy. [pdf] London: ClientEarth. Available at: <[community_power_model_legal_frameworks_june2014.pdf \(friendsoftheearth.eu\)](#)> [Accessed 26 September 2023].

Summary: Recommendations for legal frameworks that enable community power.

- Barriers:
 - Upfront investment for community projects (p.51)
 - Technical and time-consuming administrative barriers – in some countries, community power projects must obtain a license to produce (p.63)

Europe barriers drivers

OTHER ARTICLES

This is a list of articles I have not yet the time to finish and/or are outdated and/or do not have the right focus area.

2.1 JANSMA, ET AL., 2023

Note: It is hard to tell if this article I worth looking into. I had a difficult time understanding it. Very theoretical.

Jansma, S.R., Long, L.A.N. and Lee, D., 2023. Understanding Energy Citizenship: How Cultural Capital Shapes the Energy Transition. *Energies*, [e-journal] 16(2106). [https:// doi.org/10.3390/en16052106](https://doi.org/10.3390/en16052106).

- Energy citizenship: “Beyond being mere users of energy, citizens are seen as actors willing to employ social, material, and political resources in order to influence decision-making concerning energy”, like climate activism, prosumersim etc. (p. 1)
 - Literature has mainly focused on the material participation of citizens in the energy transition, like economic capital or technology, but not on the social and cultural capital/political and social dimension, which is what this study focuses on (p.2)
 - Cultural capital: “Cultural capital encompasses a range of cultural experiences and competencies, which people can deploy to secure social status and standing” (p.2)
 - Energy citizenship is a critical building block on the pathway to transforming energy systems (p.2)
 - The impact of cultural capital will vary according to the forms of energy citizenship that are in play (p.2)
 - Drivers:
 - o In relation to communicative forms of citizenship, personality characteristics such as altruism and patience also increase the likelihood of citizen participation in energy (p.4)
 - [...]
-

2.2 LAZDINS, ET AL., 2021

Lazdins, R., Mutule, A. and Zalostiba, D., 2021. PV energy communities – challenges and barriers from a consumer perspective: a literature review. *Energies*, [e-journal] 14(4873). <https://doi.org/10.3390/en14164873>.

Summary:

- Barriers for renewable energy deployment (Not specified to ECs):
 - Market failures: underinvestment in market R&D activities, low GHG emission quota prices, energy generation and utility companies' monopoly, RES solution unaffordability and high renewable market risks;
 - Informational and awareness barriers: the impact of variables and uncertainties on energy production, lack of reliable energy generation and demand data, lack of modelling solutions, lack of skilled professionals and lack of RES awareness;
 - Socio-cultural barriers: inefficient use of land for RES, lack of awareness of the benefits of using RES and weak communication between involved parties;
 - Policy barriers: institutional monopoly from existing industry and infrastructure, monopoly-based energy policy, weak intellectual property rights protection and lack of RES support activities. (Sen and Ganguly, 2017)
-

2.3 OTHER ARTICLES

(Not properly referenced)

Sen, S.; Ganguly, S. Opportunities, barriers and issues with renewable energy development—A discussion. *Renew. Sustain. Energy Rev.* 2017, 69, 1170–1181.

Brummer, V. Community energy-benefits and barriers: A comparative literature review of Community Energy in the UK, Germany and the USA, the benefits it provides for society and the barriers it faces. *Renew. Sustain. Energy Rev.* 2018, 94, 187–196. (Not the right focus area)

Lennon, B.; Dunphy, N.P.; Sanvicente, E. Community acceptability and the energy transition: A citizens' perspective. *Energy Sustain. Soc.* 2019, 9, 35.

Jasiński, J.; Kozakiewicz, M.; Sołtysik, M. Determinants of Energy Cooperatives' Development in Rural Areas—Evidence from Poland. *Energies* 2021, 14, 319. (Not the right focus area)

Wirth, S. Communities matter: Institutional preconditions for community renewable energy. *Energy Policy* 2014, 70, 236–246

Koirala, B.P.; Koliou, E.; Friege, J.; Hakvoort, R.A.; Herder, P.M. Energetic communities for community energy: A review of key issues and trends shaping integrated community energy systems. *Renew. Sustain. Energy Rev.* 2016, 56, 722–744.

Sperling, K. How does a pioneer community energy project succeed in practice? The case of the Samsø Renewable Energy Island. *Renew. Sustain. Energy Rev.* 2017, 71, 884–897.

Bauwens, T.; Gotchev, B.; Holstenkamp, L. What drives the development of community energy in Europe? The case of wind power cooperatives. *Energy Res. Soc. Sci.* 2016, 13, 136–147. (Focus on wind power development)

Ahlemeyer, et al., 2022, Success factors of citizen energy cooperatives in north western Germany: a conceptual and empirical review, *Energy, Sustainability and Society* 12:29

Maybe: Coy, Malekpour & Saeri, 2022, From little things, big things grow: Facilitating community empowerment in the energy transformation, *Energy Research & Social Sciences* 84

Palm, J., 2021. The Transposition of Energy Communities into Swedish Regulations: Overview and Critique of Emerging Regulations. *Energies*, [online] 14(16), p.4982. <https://doi.org/10.3390/en14164982>. (Probably worth looking into)

Becker, S.; Kunze, C. Transcending community energy: Collective and politically motivated projects in renewable energy (CPE) across Europe. *People Place Policy Online* 2014, 8, 180–191.

Koirala, B.P.; Araghi, Y.; Kroesen, M.; Ghorbani, A.; Hakvoort, R.A.; Herder, P.M. Trust, awareness, and independence: Insights from a socio-psychological factor analysis of citizen knowledge and participation in community energy systems. *Energy Res. Soc. Sci.* 2018, 38, 33–40.

Rahmani, S.; Murayama, T.; Nishikizawa, S. Review of Community Renewable Energy Projects: The Driving Factors and Their Continuation in the Upscaling Process. In *IOP Conference Series: Earth and Environmental Science*; IOP Publishing: Bristol, UK, 2020; p. 012033.

Ines, C.; Guilherme, P.L.; Esther, M.-G.; Swantje, G.; Stephen, H.; Lars, H. Regulatory challenges and opportunities for collective renewable energy prosumers in the EU. *Energy Policy* 2020, 138, 111212.

Walker, G. What are the barriers and incentives for community-owned means of energy production and use? *Energy Policy* 2008, 36, 4401–4405. (Probably outdated)

Models of local energy ownership and the role of local energy communities in energy transition in Europe (Commission for the environment, climate change and energy)

Förstudie: Kooperativa energilösningar -slutrapport, januari 2020 (Region Örebro län, Coompanion)

Energy communities: an overview of energy and social innovation (Caramizaru, Uihlein), 2020